SFWRENG 3MX3 2025 Midterm Solutions

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Problem 1

We know that for a discrete system that $H(\omega) = e^{-i\omega} - e^{-2i\omega}$. What is the corresponding difference equation for this system?

Solution:

We can use the fact $y(n) = H(\omega)e^{in\omega}$ when $x(n) = e^{in\omega}$ to recover the system's difference equation

$$\begin{aligned}
H(\omega) &= e^{-i\omega} - e^{-2i\omega} \\
H(\omega)e^{in\omega} &= e^{in\omega} \cdot \left(e^{-i\omega} - e^{-2i\omega}\right) \\
y(n) &= e^{-i\omega} \cdot e^{in\omega} - e^{-2i\omega} \cdot e^{in\omega} \\
&= e^{i\omega(n-1)} - e^{i\omega(n-2)} \\
&= x(n-1) - x(n-2)
\end{aligned}$$

Problem 2

What are the Fourier series coefficients of the signal $x(t) = 2 + \sin(t) + \cos(3t) + 2\cos(5t)$? You do not need to compute integrals to get full marks.

Solution:

We will rely on the fact that each term which makes up the signal is periodic, we can use the coefficients of the complex exponential expansions of sine and cosine ($sin(t) = \frac{e^{it}-e^{-it}}{2i}, \cos(t) = \frac{e^{it}-e^{-it}}{2}$) respectively) to find the Fourier series coefficients. Therefore:

$$
\begin{aligned}
X_{-5} &= 1 \\
X_{-4} &= 0 \\
X_{-3} &= \frac{1}{2} \\
X_{-2} &= 0 \\
X_{-1} &= \frac{-1}{2i} \\
X_{0} &= 2 \\
X_{1} &= \frac{i}{2i} \\
X_{2} &= 0 \\
X_{3} &= \frac{1}{2} \\
X_{4} &= 0 \\
X_{5} &= 1
\end{aligned}
$$

Notice here that:

1. The value of the Fourier series coefficient at index j corresponds to the complex exponential expansion of the term in $x(n)$ with angular frequency j.
2. Angular frequencies which do not appear in the signal are given a value of zero.
3. $X_j = \bar{X}_{-j}$

Problem 3

Define the frequency response of a continuous system.

Solution:

The frequency response, $H(\omega)$ of a continuous system $y(t)$ is defined as $y(t) = H(\omega)e^{it\omega}$ when $x(t) = e^{it\omega}$.

Problem 4

Use convolution to compute the output of a discrete system with impulse response $h(n)$ for the input $x(n) = e^{-in\omega}$. Note that the final result is the *Discrete Time Fourier Series* (DTFT)

Solution:

$$
\begin{aligned}
y(n) &= \sum_{k=-\infty}^{\infty} (h(k)x(n-k)) \\
&= \sum_{k=-\infty}^{\infty} \left(h(k)e^{i\omega(n-k)}\right) \\
&= \sum_{k=-\infty}^{\infty} \left(h(k) \cdot e^{-in\omega} \cdot e^{ik\omega}\right) \implies \\
y(n)e^{in\omega} &= \sum_{k=-\infty}^{\infty} \left(h(k) \cdot e^{ik\omega}\right) \implies \\
H(\omega) &= \sum_{k=-\infty}^{\infty} \left(h(k) \cdot e^{ik\omega}\right)
\end{aligned}
$$

Problem 5

Given the difference equation $y(n) = x(n) - y(n-2)$, compute the system's output to the signal $x(n) = 1 + \cos(n\pi + \frac{\pi}{2}) + \sin(n\frac{\pi}{4}))$ using the its frequency response.

Solution:

(I) Find the system's frequency response ($H(\omega)$).

$$y(n) = x(n) - y(n-2)$$

$$H(\omega)e^{in\omega} = e^{in\omega} - H(\omega)e^{i\omega(n-2))}$$

$$H(\omega)e^{in\omega} = e^{in\omega} - H(\omega)e^{in\omega}e^{-2i\omega}$$

$$H(\omega) = 1 - H(\omega)e^{-2i\omega}$$

$$H(\omega) + H(\omega)e^{-2i\omega} = 1$$

$$H(\omega)(1 + e^{-2i\omega}) = 1$$

$$H(\omega) = \frac{1}{1 + e^{-2i\omega}}$$

(II) Compute $H(\omega)$, $|H(\omega)|$, and $arg(H(\omega))$ for each ω in the signal $x(n)$.

ω	$H(\omega)$	$\lvert H(\omega)\rvert$	$arg(H(\omega))$
0	$1/2$	$1/2$	0
π	$1/2$	$1/2$	0
$\pi/4$	$1/(1-i)$	$\sqrt{2}/2$	$-7\pi/4$

(III) Use the formula $y(n) = |H(\omega)| \cos(n\omega + arg(H(\omega)))$ for each of the terms we computed.

$$
\begin{aligned}
y(n) &= |H(\omega_1)| \cos(n\omega_1 + arg(H(\omega_1))) + |H(\omega_2)| \cos(n\omega_2 + arg(H(\omega_2))) \\
&+ |H(\omega_3)| \cos(n\omega_3 + arg(H(\omega_3))) \\
&= \frac{1}{2}\cos(0n + 0) + \frac{1}{2}\cos(n\pi + 0) + \frac{\sqrt{2}}{2}\cos(\frac{pi}{4} + \frac{-7\pi}{4}) \\
&= \frac{1}{2} + \frac{1}{2}\cos(\pi) + \frac{\sqrt{2}}{2}\cos(\frac{\pi}{4} + \frac{-7\pi}{4})
\end{aligned}
$$

Problem 6

Compute the impulse response and the frequency response of $y(n) = x(n-3)$.

(I) Compute the impulse response ($h(n)$).

Substituting $x(n) = \delta(n)$, we find:

$$
\begin{aligned}
y(n) &= x(n-3) \implies \\
h(n) &= \delta(n-3)
\end{aligned}
$$

(II) Compute the frequency response ($H(\omega)$).

Using $y(n) = H(\omega)e^{in\omega}$ when $x(n) = e^{in\omega}$, we find:

$$
\begin{aligned}
y(n) &= x(n-3) \\
H(\omega)e^{in\omega} &= e^{i\omega(n-3)} \\
H(\omega)e^{in\omega} &= e^{in\omega} \cdot e^{-3i\omega} \\
H(\omega) &= e^{-3i\omega}
\end{aligned}
$$

Problem 7

What is the output y(n) given the system $y(n) = x(n) + \alpha y(n-1)$ and input $x(n) = 1 \ \forall n \in \mathbb{N}$?

Solution:

$$\begin{gathered} y(0) = 1 + \alpha(0) = 1 \\ y(1) = 1 + \alpha(1) = 1 + \alpha \\ y(2) = 1 + \alpha(1 + \alpha) = 1 + \alpha + \alpha^2 \\ y(3) = 1 + \alpha(1 + \alpha + \alpha^2) = 1 + \alpha + \alpha^2 + \alpha^3 \\ \vdots \end{gathered}$$

Extrapolating, we can surmise that:

$$y(n) = 1 + \alpha + \alpha^2 + \ldots + \alpha^n$$

Problem 8

The Fourier series for the 4-periodic signal $x(n) = \sin(\frac{\pi}{2}n)$ was computed and yielded:

$$X_0 = 0, \ X_1 = -i, \ X_2 = 0, \ X_3 = i$$

Compute the values of x(n) for one period.

Solution:

(I) Compute ω_0.

Since we are given that $x(n)$ is 4-periodic, we can use the formula $\omega_0 = \frac{2\pi}{P}$ to find that:

$$\begin{aligned} \omega_0 &= \frac{2\pi}{4} \\ &= \frac{\pi}{2} \end{aligned}$$

(II) Compute the values of $x(n)$ for one period.

Using the formula:

$$x(n) = \sum_{k=0}^{p=1} \left(X_k e^{ink\omega_0} \right)$$

we can compute the values of x(n) as follows:

$$\begin{aligned}
x(0) &= \sum_{k=0}^{3} \left(X_k e^{i(0)k\frac{\pi}{2}} \right) = (0 - i + 0 - i) = 0 \\
x(1) &= \sum_{k=0}^{3} \left(X_k e^{i(1)k\frac{\pi}{2}} \right) = (-ie^{\frac{\pi}{2}i} + ie^{\frac{3\pi}{2}i}) = 2 \\
x(2) &= \sum_{k=0}^{3} \left(X_k e^{i(2)k\frac{\pi}{2}} \right) = (-ie^{i\pi} + ie^{3i\pi}) = 0 \\
x(3) &= \sum_{k=0}^{3} \left(X_k e^{i(3)k\frac{\pi}{2}} \right) = (-ie^{\frac{3\pi}{2}i} + ie^{\frac{9\pi}{2}i}) = -2
\end{aligned}$$

Problem 9

Give the difference equation of a minimum order causal filter that filters the discrete frequencies π and $\frac{\pi}{2}$ and has a gain of 1 at DC. Derive the frequency response and corresponding difference equation of such a filter.

Solution:

(I) Find a prototype filter $\tilde{H}(\omega)$.

We know that our prototype filter must satisfy $|\tilde{H}(0)| = 1$ and $\tilde{H}(\pi))$, $\tilde{H}(\frac{\pi}{2}) = 0$. Using this we can arrive at the following prototype filter.

$$\begin{aligned}
\tilde{H}(\omega) &= (e^{-i\omega} - e^{-i\pi})(e^{-i\omega} - e^{-\frac{\pi}{2}i})(e^{-i\omega} - e^{\frac{\pi}{2}}) \\
&= (e^{-i\omega} + 1)(e^{-i\omega} - i)(e^{-i\omega} + i)
\end{aligned}$$

Note here that to ensure conjugate complex symmetry at $\frac{\pi}{2}$, we had to include both $\frac{\pi}{2}$ and $\frac{-\pi}{2}$ as zeros of our prototpe filter.

Evaluating for $\omega = 0$, which is the DC, we can see the gain of our filter is:

$$\begin{aligned} |\tilde{H}(0)| &= |(1+1)(1-i)(1+i)| \\ &= 2(1-i^2) \\ &= 2(1-(-1)) \\ &= 4 \end{aligned}$$

Thus, our actual filter, $H(\omega)$, should be given by:

$$\begin{aligned} H(\omega) &= \frac{1}{4}(e^{-i\omega}+1)(e^{-i\omega}-i)(e^{-i\omega}+i) \\ &= \frac{1}{4}(e^{-i\omega}+1)(e^{-2i\omega}+ie^{-i\omega}-ie^{-i\omega}+1) \\ &= \frac{1}{4}(e^{-i\omega}+1)(e^{-2i\omega}+1) \\ &= \frac{1}{4}(e^{-3i\omega}+e^{-2i\omega}+e^{-i\omega}+1) \end{aligned}$$

(II) Find the filter's difference equation.

Using $y(n) = H(\omega)e^{in\omega}$ when $x(n) = e^{in\omega}$, we find:

$$\begin{aligned} H(\omega) &= \frac{1}{4}(e^{-3i\omega}+e^{-2i\omega}+e^{-i\omega}+1) \\ H(\omega)e^{in\omega} &= \frac{e^{in\omega}}{4}(e^{-3i\omega}+e^{-2i\omega}+e^{-i\omega}+1) \\ H(\omega)e^{in\omega} &= \frac{1}{4}(e^{-3i\omega}e^{in\omega}+e^{-2i\omega}e^{in\omega}+e^{-i\omega}e^{in\omega}+e^{in\omega}) \\ H(\omega)e^{in\omega} &= \frac{1}{4}(e^{-3i\omega(n-3)}+e^{-2i\omega(n-2)}+e^{-i\omega(n-1)}+e^{in\omega}) \\ y(n) &= \frac{1}{4}(x(n-3)+x(n-2)+x(n-1)+x(n)) \end{aligned}$$

Problem 10

Given the following matricies:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, C = \begin{pmatrix} 1 & 1 \end{pmatrix}, D = \begin{pmatrix} 0 \end{pmatrix}$$

Compute the zero state output to the input $x(n) = \delta(n)$. (Show $s(n)$ and $y(n)$ for $n = 0, 1, 2,$ and 3.

Solution:

Using the state space equations, we can compute the outputs of the system as follows:

$\underline{S(n)}$

$$S(0) = 0$$
$$S(1) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} S(0) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$S(2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} S(1) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(1) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
$$S(3) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} S(2) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(2) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$\underline{y(n)}$

$$y(0) = \begin{pmatrix} 1 & 1 \end{pmatrix} (0) = 0$$
$$y(1) \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1$$
$$y(2) \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1$$
$$y(3) \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1$$